USER GUIDE

Jacon PPS Report

A complete walkthrough of the application: loading a post-processed scan, navigating and measuring in 3D, extracting segments, annotating findings, running the thickness analysis, and exporting a professional PDF report.

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▲ WARNING

To prevent serious injury and machine damage, only trained personnel should install replacement parts.

Only genuine replacement parts should be used to repair Jacon machinery.

Failure to do so could result in personal injury or equipment damage.

A copy of this parts manual should be kept in the site maintenance repair document library.

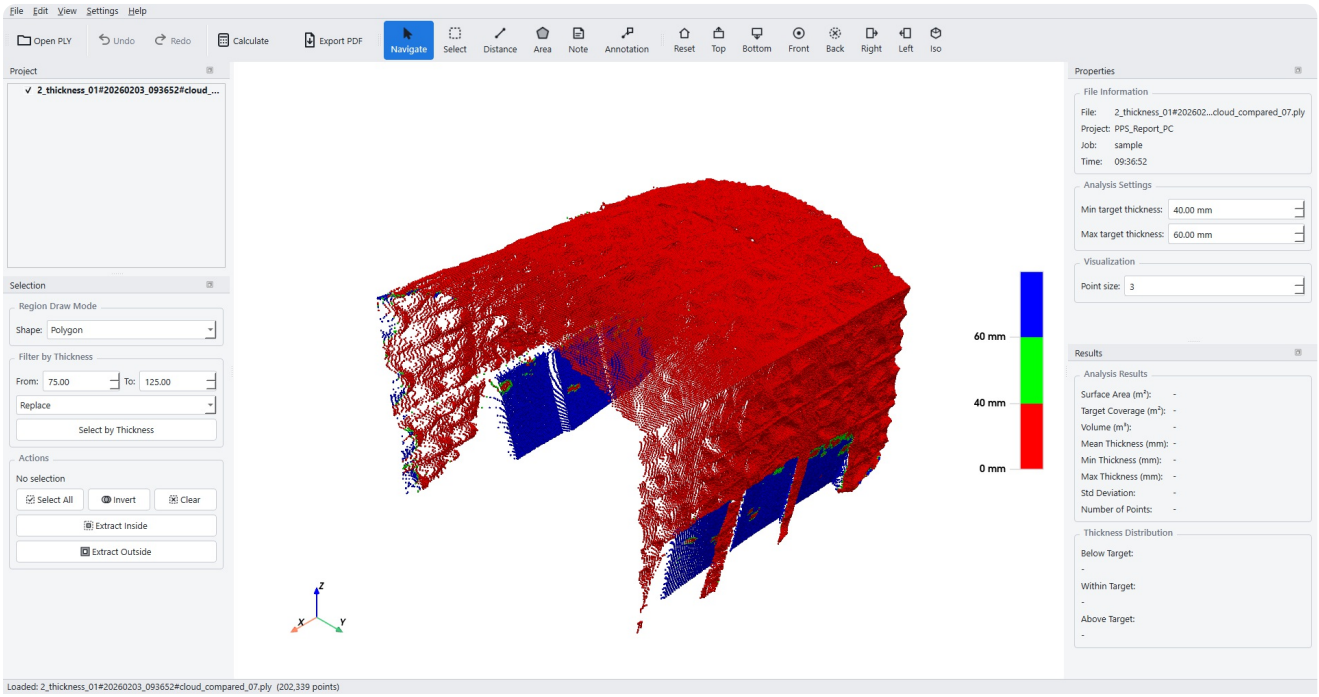
1 Overview

Jacon PPS Report reads a 3D point cloud (`.ply`) produced by the **Jacon Pre-Post Scanner (PPS) System**, which scans a tunnel or mine face **before** shotcrete is sprayed (the *Pre-Scan*, the raw excavated rock surface) and again **after** spraying (the *Post-Scan*). The PPS System computes the perpendicular distance between the Post-Scan and Pre-Scan surfaces at every point and writes that thickness value into the `.ply` file.

Jacon PPS Report does **not** perform that Pre-Scan/Post-Scan comparison itself — its job is post-processing and reporting on data that already carries per-point thickness. It colors the cloud by thickness, lets you inspect it interactively in 3D, isolate regions of interest as separate segments, measure distances and areas, annotate findings directly on the model, compute summary statistics against a target thickness range, and produce a PDF report suitable for handing to a client or QA reviewer.

Typical end-to-end workflow:

1. Open the post-processed `.ply` scan file (already carries per-point thickness from the PPS System).
2. Review / adjust the target thickness range (Min / Max).
3. Optionally select and extract one or more segments of interest.
4. Measure distances or areas, and place Notes/Annotations to document findings.
5. Run **Calculate** to get the full statistical breakdown.
6. Export the PDF report (and optionally save the working session as a project file).



The main window: tool toolbar across the top-middle, the **Project** and **Selection** docks on the left, **Properties** and **Results** on the right, and the 3D viewport in the center with the thickness color legend.

2 Interface Tour

The main window is built from a menu bar, two toolbars, a central 3D viewport, and four dockable panels:

AREA	PURPOSE
Main toolbar	Open PLY, Undo/Redo, Calculate, Export PDF
Tool toolbar	The six interaction tools — Navigate, Select, Distance, Area, Note, Annotation — plus quick view-orientation buttons (Reset, Top, Bottom, Front, Back, Right, Left, Iso)
Project dock	Tree of segments (layers) with their notes/annotations/measurements nested underneath; visibility, rename, delete, edit and move all happen here
Selection dock	Region-select shape, thickness filter, and segment extraction/crop actions
Properties dock	Loaded file information, Target Min/Max thickness, default point size
Results dock	Calculation output: surface area, volume, thickness statistics, and the below/within/above-target distribution

Every dock can be dragged to a different edge of the window, floated as its own window, or hidden/shown from the `View` menu. If the layout ever gets into an awkward state, use `View → Reset Layout` to restore the default arrangement.

3 Opening a PLY File

Use `File → Open PLY…` (`Ctrl + O`) or the **Open PLY** button on the main toolbar, and choose a `.ply` file that already contains per-point thickness values computed by the PPS System (the Post-Scan surface compared against the Pre-Scan rough surface — not a CAD design profile).

Filename & folder convention

If the file lives under a path shaped like `…/Projects/<ProjectName>/<JobNumber>/<JobNumber>#<yyyyMMdd_hhmmss>#<SegmentName>.ply` , the project name, job number, scan timestamp and segment name are parsed automatically and shown in the **Properties** dock, and reused to build the default PDF report filename. If the path doesn't match this convention, the file still loads normally — the project/job fields simply fall back to the folder and file names as-is.

Default target range

The initial **Target Min/Max** thickness is resolved automatically:

- If a `job_info.json` file sits next to the PLY with a `parameters.target_thickness` and `parameters.tolerance` , the range is set to *target − tolerance* to *target + tolerance*.
- Otherwise it defaults to **40–60 mm**.

You can always adjust Min/Max afterwards in the Properties dock — see Calculation & Targets.

4 Saving & Opening Projects

Besides the source `.ply`, the application can save your entire working session — segments, notes, annotations, measurements, target range, camera position, point size — to a `.ppsproj` project file, so you can close the app and pick up exactly where you left off.

ACTION	SHORTCUT	NOTES
Save Project	`Ctrl + S`	Saves to the current project path; behaves like Save As on the first save
Save Project As…	`Ctrl + Shift + S`	Always prompts for a location
Open Project…	`Ctrl + Shift + O`	Loads a `.ppsproj` file
Recent Projects	—	Submenu under `File` listing recently opened projects

> **Unsaved changes:** opening another file/project, or closing the app, while there are unsaved changes prompts you to Save, Discard, or Cancel — nothing is lost silently.

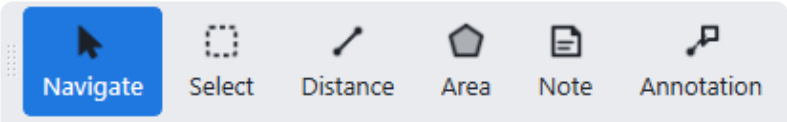
Once a project has been saved, exporting a PDF report from it works immediately without needing to reload the original `.ply` file.

5 3D Navigation & Tools

Camera controls

Left-drag to orbit, right-drag or scroll to zoom, middle-drag to pan. The view-orientation buttons on the right side of the tool toolbar (**Reset, Top, Bottom, Front, Back, Right, Left, Iso**) snap the camera to standard viewpoints.

The six tools



Tool toolbar: Navigate, Select, Distance, Area, Note, Annotation.

TOOL	SHORTCUT	WHAT IT DOES
Navigate	`V / 1`	Orbit/pan/zoom the camera; select, drag-move, right-click edit/delete any existing Note, Annotation, or measurement
Select	`S / 2`	Draw a selection region (Polygon / Rectangle / Lasso) on the cloud, used to extract or crop segments
Distance	`D / 3`	Measure the straight-line distance between two points on the cloud
Area	`A / 4`	Draw a region and compute its true surface area
Note	`N / 5`	Place free-floating 2D text anywhere on the view, with no 3D anchor
Annotation	`G / 6`	Place text anchored to a specific point on the cloud, connected by a leader line

> **Auto-return to Navigate:** as soon as a creation tool finishes its action — a Note/Annotation is placed, a Distance/Area measurement completes, or a segment is extracted/cropped — the active tool switches back to **Navigate** automatically, so you can immediately interact with what you just created without pressing another button.

Press `Esc` at any time to cancel an in-progress action (e.g. a half-drawn polygon); pressing it again while idle returns to Navigate.

6 Selection & Segments

Drawing a selection

Switch to the **Select** tool and choose a shape in the Selection dock:

- **Polygon** — left-click to add each vertex, then double-click, right-click, or press `Enter` to close the shape (needs at least 3 points). `Backspace` removes the last point while drawing.
- **Rectangle** — click and drag to define the box.
- **Lasso** — click and drag to freehand-draw the outline.

Hold `Shift` while closing the shape to *add* to the current selection, or `Ctrl` to *subtract* from it; with neither held, the new shape replaces the current selection.

Filtering by thickness

The *Filter by Thickness* panel lets you type a thickness range and click **Select by Thickness** to automatically select every point whose value falls inside it — useful for isolating under- or over-sprayed areas without manually drawing a shape.

Extracting a segment

Once you have a selection, two actions turn it into a new segment (layer):

- **Extract Inside** — creates a new segment from the points *inside* the current selection.
- **Extract Outside** — creates a new segment from the points *outside* the current selection (crop away what's selected).

The other Actions buttons — **Select All**, **Invert**, and **Clear** — operate on the current selection without creating a segment.

A newly extracted segment is shown on its own (every other layer is temporarily hidden) and appears as a new top-level entry in the Project tree. Notes, Annotations and measurements placed on a segment belong to it and will hide/show together with it.

7 Measuring Distance & Area

Distance

Select the **Distance** tool, click the first point on the cloud, move the mouse to see a live preview line and running distance, then click the second point to finish. The result is attached to the nearest segment and appears in the Project tree.

Area

Select the **Area** tool and draw a frame on the cloud the same way as a selection shape (polygon, rectangle or lasso). When the frame is closed:

- It **snaps onto the cloud surface you are looking at** — the outline is made of cloud points, so it stays attached to the cloud and rotates/zooms with it.
- Only the points **enclosed by that outline on that surface** are measured. Surfaces hidden behind the region (for example the opposite tunnel wall) are never included, whatever the viewing angle.
- The true surface area (not the flat projected area) of those points is computed using Ball-Pivoting surface reconstruction (BPA) — the same method the PDF report itself uses — and is displayed on the model a few seconds later, once the background calculation finishes.

Draw the frame on the cloud itself: parts of the frame drawn on empty background are ignored, and a frame entirely off the cloud creates no measurement. Area measurements from projects saved with an older version keep their value and label but have no outline.

Changing a measurement's color and font size

Each Distance and Area measurement has its own color (used for its label and its line/outline) and label font size. With the **Navigate** tool active, right-click the measurement's label and choose **Edit…** — or right-click (or double-click) it in the Project tree — then pick a color and font size. The change can be undone with `Ctrl + Z` and is saved in the project file.

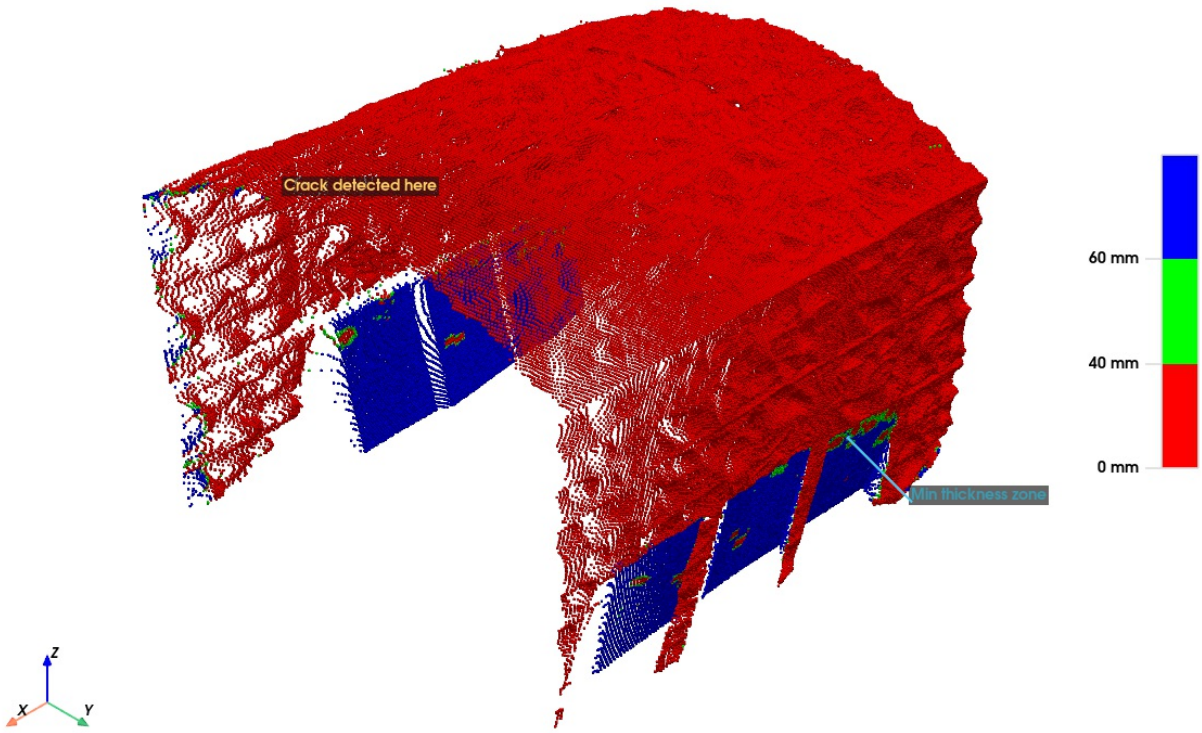
8 Notes & Annotations

Text placed directly on the 3D view, exported together with the PDF report

Creating one

Note (`N`): click anywhere on the view to type text right on the spot. A Note has no 3D anchor — it stays fixed on screen regardless of how the camera moves, and attaches to whichever segment is currently visible (or the original cloud if no segment is visible) purely for organization in the Project tree.

Annotation (`G`): click a point on the cloud — one end is pinned to that 3D point (marked with a small colored sphere), the other end is the text label, joined by a leader line. An Annotation tracks the model as the camera orbits, and is attached to whichever segment is spatially closest to the clicked point.



Example: "Crack detected here" is a screen-fixed **Note**; "Min thickness zone" is an **Annotation** anchored to the model with a leader line.

Editing, moving, and deleting

Switch to the **Navigate** tool (`v`) to work with existing Notes/Annotations:

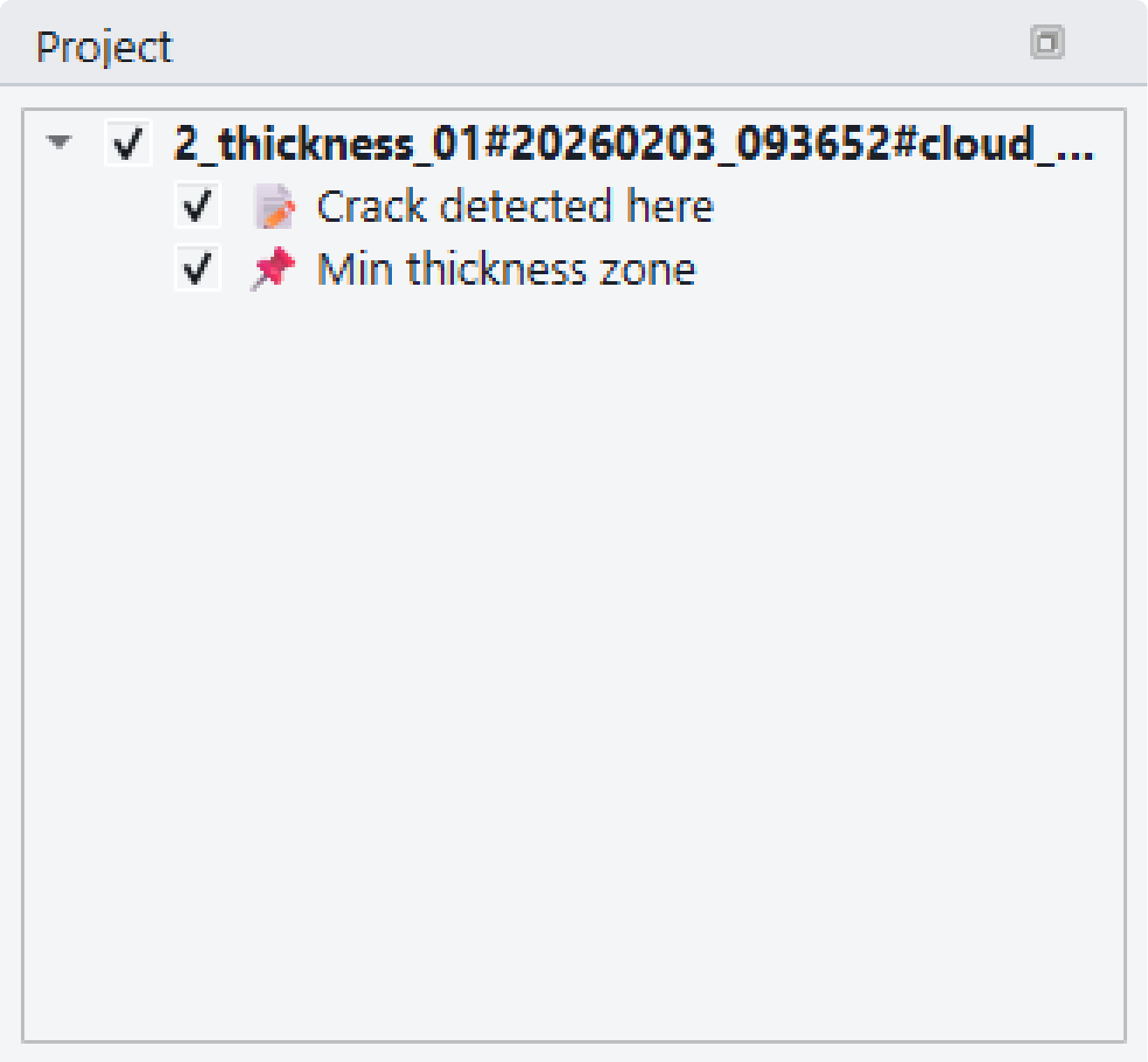
- Click directly on a Note/Annotation in the 3D view to select it (it highlights in orange), or select it from the Project tree — either way stays in sync with the other.
- Drag it to move the text to a new position; it tracks live under the cursor while dragging.
- Right-click for a context menu with **Edit…** (change the text/color/font size) and **Delete**.
- Press `Delete` to remove whichever Note/Annotation/measurement is currently selected.

The same Edit/Move/Delete actions are also available by right-clicking the item directly in the Project tree.

9 The Project Tree

The **Project** dock on the left shows every segment (layer) as a top-level row, with its Notes, Annotations and measurements nested underneath as child rows. Toggling a segment's checkbox hides or shows everything nested inside it in one action.

Selecting multiple rows (`Ctrl` / `Shift` -click) enables a bulk **Delete** action, applied as a single undo step. Notes/measurements with no owning segment (for example, after their segment was deleted) are grouped under an **(Unassigned)** entry rather than being lost.



Project tree, checkboxes control visibility; right-click a segment to Rename/Delete, or an object to Edit/Move/Hide/Delete.

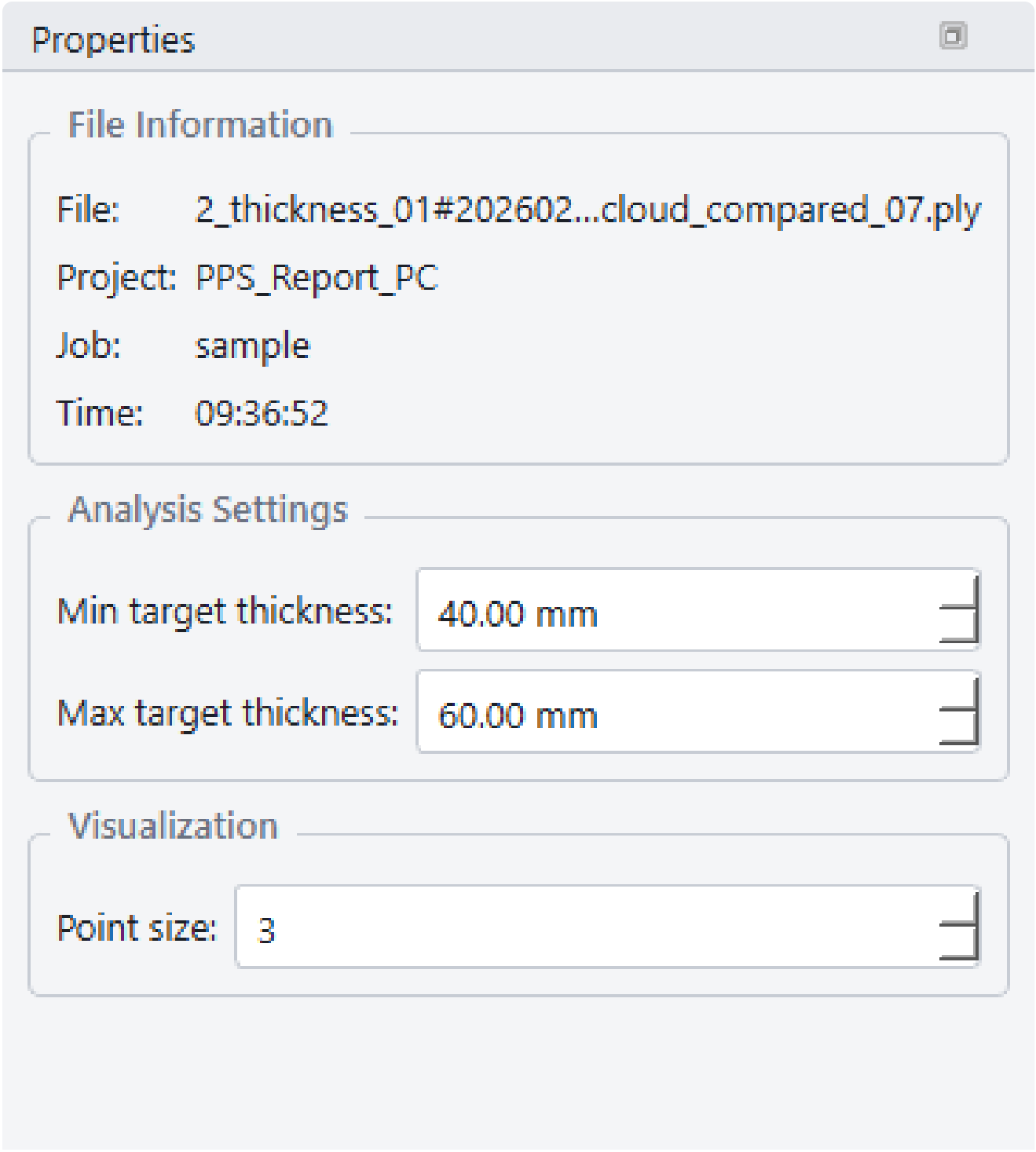
ROW TYPE	RIGHT-CLICK MENU
Segment (layer)	Rename..., Delete layer (not available for the original cloud)
Note / Annotation	Edit..., Move, Hide/Show, Delete
Distance / Area measurement	Edit... (color and font size), Hide/Show, Delete — double-click also opens Edit...

10 Calculation & Targets

In the **Properties** dock, set **Min target thickness** and **Max target thickness** (mm) to match the design requirement. The model re-colors automatically about 0.5 seconds after you stop typing. Max is always kept ≥ Min automatically.

Click **Calculate** on the main toolbar to run the full analysis of the **visible** clouds. This runs in the background so the UI stays responsive on large clouds.

The results belong to the targets and visible clouds they were calculated for. Changing either — editing Min/Max, or showing/hiding a cloud in the Project tree — clears the Results dock; click **Calculate** again (renaming or recoloring a cloud does not).



Properties dock, file info, Target Min/Max, point size.

Reading the Results

FIELD	MEANING
Surface Area (m^2)	Total reconstructed surface area of the visible cloud
Target Coverage (m^2)	Surface area of the region that reached the minimum target thickness
Volume (m^3)	Target-coverage area × mean thickness of that region
Mean / Min / Max Thickness (mm)	Statistics computed over points that reached the target thickness
Std Deviation	Standard deviation of thickness over the same set of points
Number of Points	Total point count in the visible cloud

FIELD	MEANING
Below / Within / Above Target	Point counts and percentages split by the Target Min/Max range

The three color classes correspond to the colors configured in Settings:

- **Below Target Min** (default: red) — insufficient shotcrete thickness.
- **Within Target Min–Max** (default: green) — meets the design requirement.
- **Above Target Max** (default: blue) — over-sprayed beyond the target.

Note: thickness values with an absolute magnitude below 12 mm are treated as no measurable deviation, and values above 500 mm are treated as scan noise — both are excluded from the "reached target" statistics. The PPS System hardware's own valid measurement range is 0–300 mm, so the 500 mm cutoff is a deliberately generous software-side margin above that — it only catches genuine scan artifacts (e.g. reflections), not real thickness readings.

11 Exporting PDF Reports

Once **Calculate** has run, click **Export PDF...** (`Ctrl + E`) on the toolbar and choose where to save. If there are no up-to-date results — never calculated, or the targets or visible clouds changed since — a **Calculation required** prompt appears: **OK** runs Calculate and then continues straight to saving the report, **Cancel** stops the export. The default filename is built from the parsed project name, job number, scan time and segment name (see Opening a PLY File).

The generated report includes:

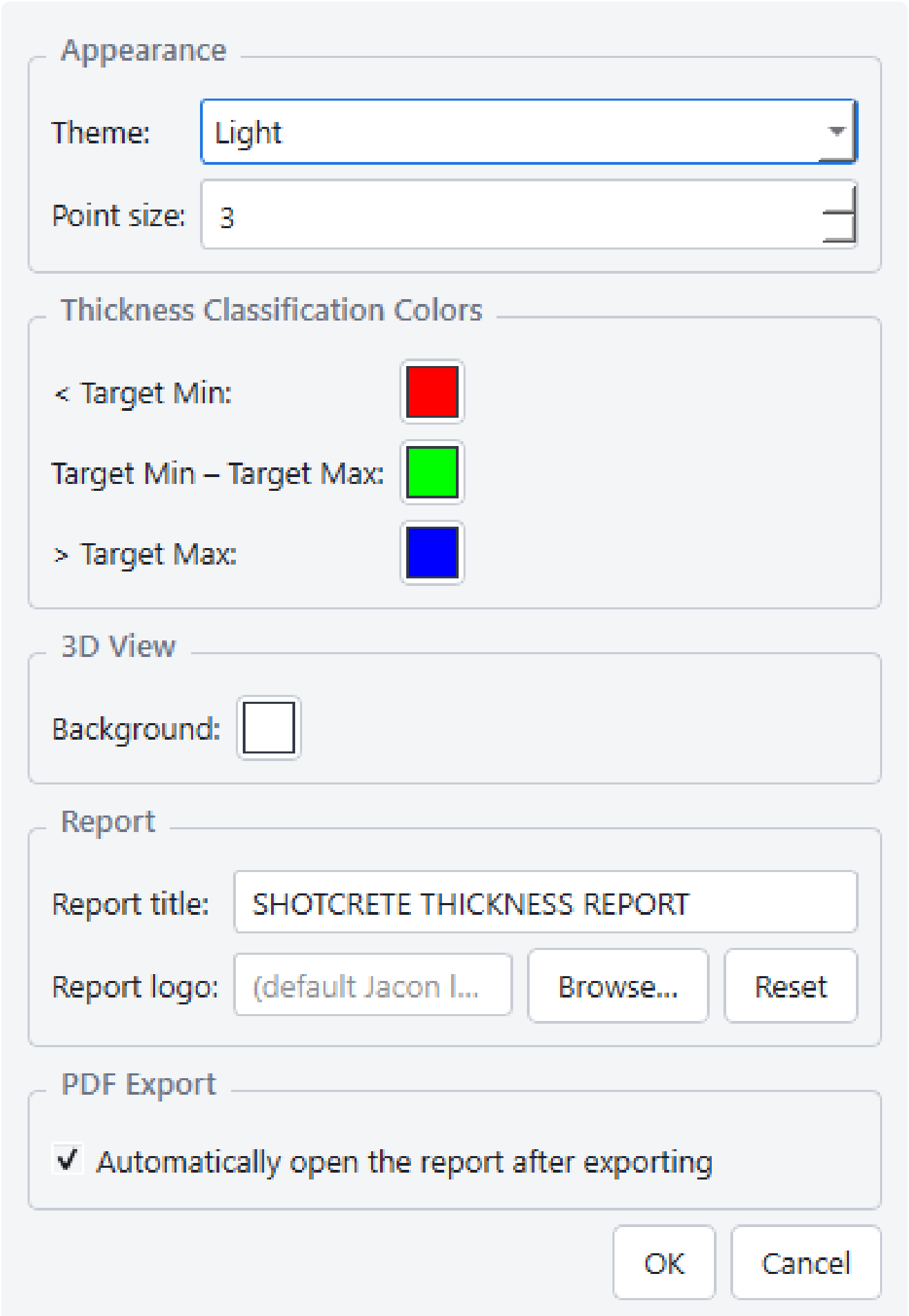
- A snapshot of the current 3D view (including any visible Notes/Annotations and the color legend).
- The full Calculate results table and thickness distribution.
- Project/job information parsed from the filename.
- The set of segments that were visible at export time.

Automatically opening the report: go to `Settings → Preferences...` , section *PDF Export*, and toggle **"Automatically open the report after exporting"**. When enabled (the default), the PDF opens in your system's default PDF viewer immediately after export completes — no confirmation dialog is shown either way.

Customizing the report title and logo: go to `Settings → Preferences...` , section *Report*. Change **Report title** to replace the heading printed at the top of every exported PDF, and use **Report logo → Browse...** to pick your own image (PNG/JPG/SVG) instead of the default Jacon logo — click **Reset** to go back to the default. Both settings are saved per-user and apply to every report exported afterwards.

12 Application Settings

Open `Settings → Preferences...` to configure theme, point size, thickness classification colors, 3D view background, report title/logo, and PDF auto-open. These preferences are stored per-user and persist across sessions and across different point cloud files.



The Settings dialog, including the PDF Export auto-open option.

SETTING	EFFECT
Theme	Light or Dark UI appearance
Point size	Rendered point size. Also adjustable in the Properties dock; the last size used is remembered for the next launch, and opening a project applies the size saved in it
Thickness Classification Colors	The three colors used for Below / Within / Above target
3D View Background	Viewport background color — independent of Theme, since it is baked into the PDF report's screenshot
Report title	The heading printed at the top of the PDF report (default: "SHOTCRETE THICKNESS REPORT")
Report logo	Custom logo image (PNG/JPG/SVG) printed on the report; **Reset** reverts to the default Jacon logo
Automatically open the report after exporting	Whether Export PDF opens the resulting file automatically (see above)

13 Undo, Redo & Safety

Nearly every change — selections turned into segments, renames, deletes, target range edits, moving/editing a Note or Annotation — is recorded on a single undo stack, so `Ctrl + Z` / `Ctrl + Y` (Undo/Redo) works consistently across the whole application. Bulk operations (deleting multiple selected rows in the Project tree, for example) are grouped into a single undo step.

The window title and the Save Project action reflect whether there are unsaved changes; closing the app or opening another file/project while changes are pending will always ask whether to save first.

14 Keyboard Shortcuts

Tools

SHORTCUT	TOOL
`V / 1`	Navigate
`S / 2`	Select
`D / 3`	Distance
`A / 4`	Area
`N / 5`	Note
`G / 6`	Annotation

General

SHORTCUT	ACTION
`Esc`	Cancel the in-progress action; press again to return to Navigate
`Delete`	Delete the currently selected Note/Annotation/measurement (Navigate tool)
`Ctrl + Z / Ctrl + Y`	Undo / Redo
`Ctrl + O`	Open PLY...
`Ctrl + S`	Save Project
`Ctrl + Shift + S`	Save Project As...
`Ctrl + Shift + O`	Open Project...
`Ctrl + E`	Export PDF...
`Ctrl + A`	Select All (every visible point)
`Ctrl + Q`	Exit

15 Tips & Troubleshooting

MESSAGE / SITUATION	WHAT IT MEANS
"Calculation required" when exporting	The results are missing or out of date (targets or visible clouds changed). Press OK to calculate and continue the export.
"Please load a PLY file first!"	Shown by Calculate/Export when no cloud is loaded yet.
A tool button is disabled	All six tools require a point cloud to be loaded; they re-enable automatically once one is.
Segments look overlapped / hidden	Extracting a segment automatically hides every other layer — check the Project tree checkboxes if something seems missing.
An object I deleted a segment for is gone	It hasn't been deleted — it moved to the **(Unassigned)** group in the Project tree.
Nothing happens when I click a Note	Selecting/moving/editing existing objects only works while the **Navigate** tool is active, not while Note/Annotation is active.

For anything not covered here, check `Help → About` for the application version, which is useful information when reporting an issue.

Jacon Contact List

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